SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SUBJECT: DATA WAREHOUSING AND DATA MINING | COURSE CODE: CSA1604

OLAP & Data Cubes

Analysing data from different perspectives

TECHNICAL PRESENTATION | CO1 | ASSESSMENT TOOL 2

STUDENT DETAILS

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Code of Conduct: Integrity in Work • Professionalism in Practice • Safety First • Respect for All

A sales database records every order, but it cannot answer the manager

WHAT THE DATABASE STORES (OLTP)

Order	Product	City	Date	Amount
#10231	Laptop	Chennai	03-Jan	₹62,000
#10232	Phone	Chennai	03-Jan	₹18,500
#10233	Phone	Bangalore	04-Jan	₹21,000
#10234	Tablet	Hyderabad	04-Jan	₹27,000
… and millions more rows, one per sale				

WHAT THE MANAGER ASKS

Which product sold most in Chennai in February? — Product, Location, Time

How did February compare with January? — Time

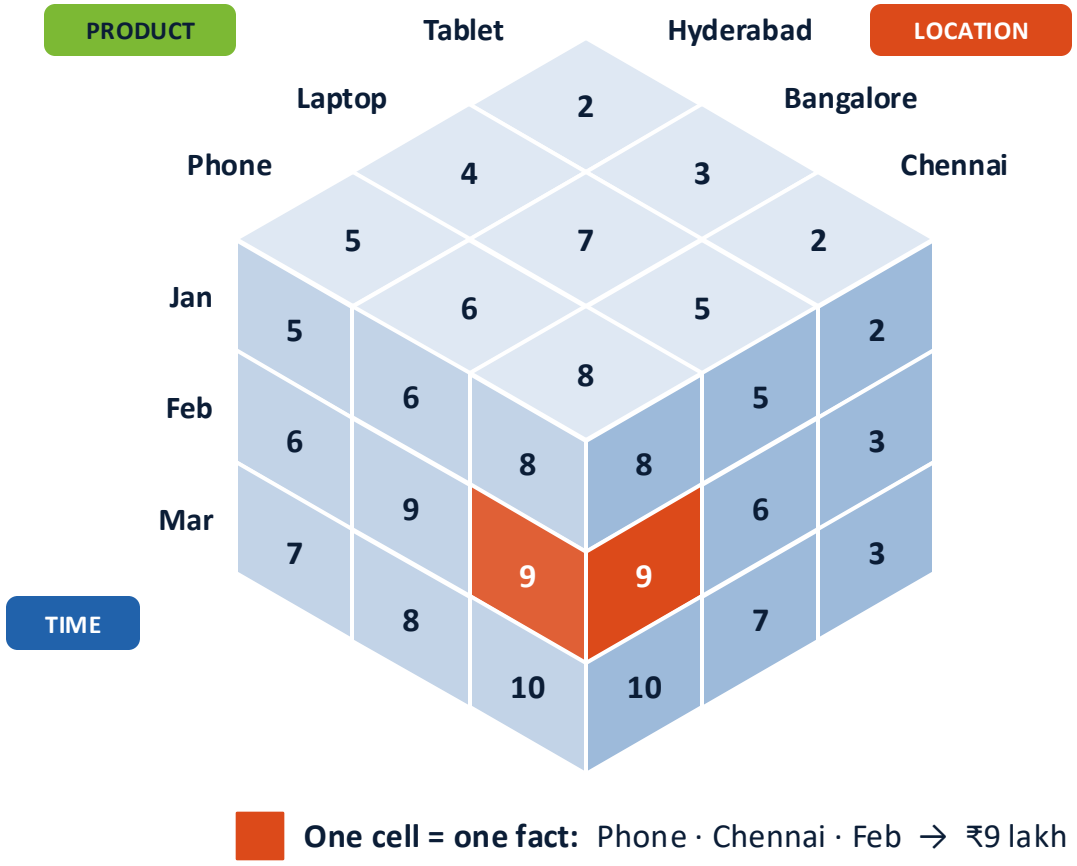
Which city generated the highest sales? — Location

OLAP Online Analytical Processing

Interactive analysis of summarised, historical data along several dimensions at once. Built for questions, not for transactions.

	OLTP: runs the business	OLAP: analyses the business
Typical work	Insert or update one order	**Aggregate millions of rows**
Data	Current, detailed	**Historical, summarised**
Design	Normalised ER tables	**Star / snowflake schema**
Users	Clerks, apps, customers	**Analysts, managers**
Measured by	Transactions per second	**Query response time**

A data cube stores one measure at every combination of dimensions



One cell = one fact: Phone · Chennai · Feb → ₹9 lakh

Dimension

A perspective to analyse by. Here: Product, Location, Time.

Measure

The number in each cell. Here: sales in ₹ lakh, aggregated with SUM.

Concept hierarchy

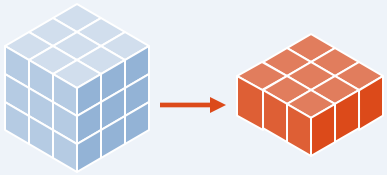
Levels of detail inside one dimension:

TIME	Day → Month → Quarter → Year
LOCATION	City → State → Country
PRODUCT	Item → Category

Stored as a star schema



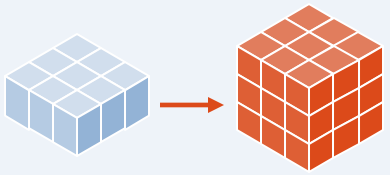
Five operations let the analyst move around the cube



Roll-up

Climb a hierarchy or drop a dimension: detail becomes summary.

Month → Quarter
Phone, Chennai: 8 + 9 + 10 = ₹27 L



Drill-down

Step down a hierarchy: summary opens into detail.

Quarter → Month
Laptop ₹56 L = 16 + 19 + 21



Slice

Fix one dimension to a single value to get a 2-D view.

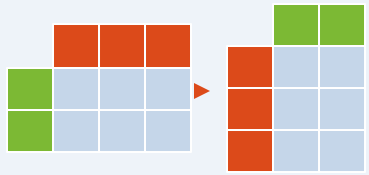
Month = Feb
Product × City table, ₹51 L in all



Dice

Pick values on two or more dimensions to get a sub-cube.

{Laptop, Phone} × {Chennai, Bangalore} × {Jan, Feb} = ₹58 L



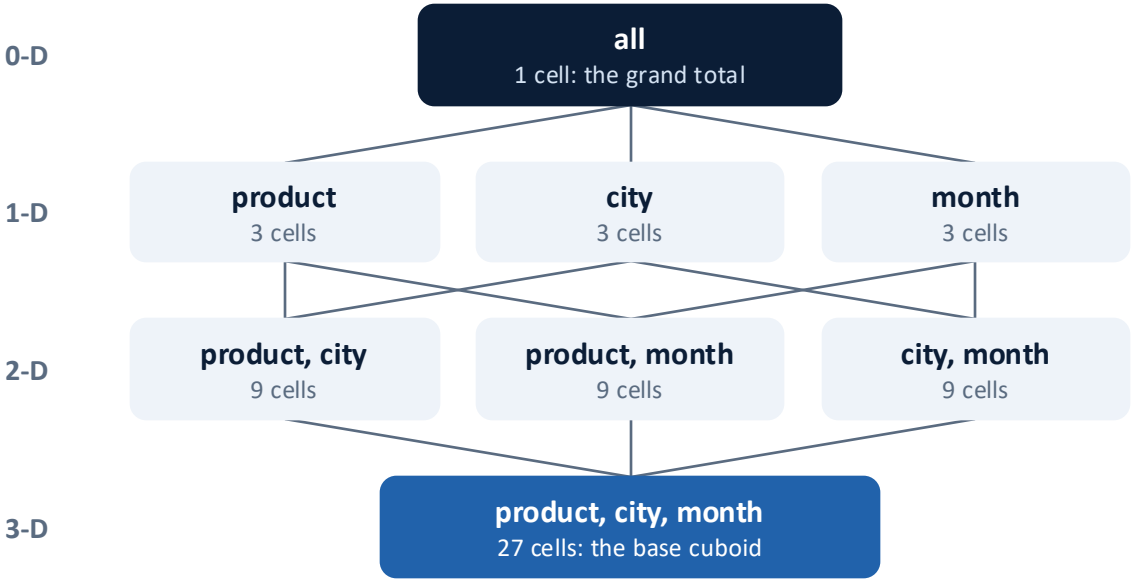
Pivot

Rotate the axes to see the same cells from another angle.

Product × City becomes City × Product

QUICK CHECK **"Total phone sales in Chennai for the whole quarter?"** Which two operations get you there?

n dimensions give 2^n cuboids, so the warehouse chooses what to precompute



Moving up the lattice is a roll-up; moving down is a drill-down. 8 cuboids, 64 cells.

WHERE THE CUBE IS STORED

ROLAP
Relational tables and SQL; scales to very large data

MOLAP
Multidimensional arrays; fastest queries

HOLAP
Detail in ROLAP, summaries in MOLAP

2^n cuboids for n dimensions

3 dimensions → 8 10 dimensions → 1,024

With hierarchies: $\prod (L_i + 1)$. 10 dimensions of 4 levels each: $5^{10} \approx 9.8$ million

WHAT TO PRECOMPUTE

None
Aggregate on demand. No extra storage, slow queries.

Full
All cuboids. Fastest queries, most storage.

Partial
Only useful cuboids, e.g. an iceberg cube. The practical choice.

HOW IT IS COMPUTED

MultiWay array aggregation full cube from array chunks; suits MOLAP

BUC iceberg cube; prunes cells below minimum support (Apriori)

Star-Cubing combines both ideas using star-tree structures

OLAP turns stored data into answers, from any angle

1 **Model**
A data cube organises a measure by dimensions and their hierarchies.

2 **Navigate**
Roll-up, drill-down, slice, dice and pivot change the view, never the data.

3 **Engineer**
2^n cuboids force a choice: what to precompute and where to store it.

WHERE IT IS USED

Retail
Sales by product, store and season

Banking
Risk and fraud trends by branch and time

Healthcare
Bed occupancy by ward and month

Education
Results by department and semester

Responsible use
Small groups can still expose individuals after aggregation. Limit how far users can drill down and apply role-based access.

Thank you. Questions?

REFERENCES

[1] J. Han, M. Kamber, J. Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann, 2011 (Ch. 4 and 5).
[2] A. Berson, S. J. Smith, Data Warehousing, Data Mining & OLAP, Tata McGraw-Hill, 2007.
[3] J. Gray et al., "Data Cube: A Relational Aggregation Operator Generalizing Group-By, Cross-Tab, and Sub-Totals," Data Mining and Knowledge Discovery 1(1), 1997.


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